



Soroush Haghi

Computer Engineering | Software & AI | Quantum Technologies

Braunschweig / Münster, Germany

s.haghi.career@outlook.com | linkedin.com/sorousshaghi

sorousshaghi.github.io



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PROFESSIONAL PROFILE

M.Sc. student in Quantum Technologies at TU Braunschweig with a B.Sc. in Computer Engineering (AI specialization). Practical work is strongest in Python-based machine learning and computer vision, sensor and signal processing, MATLAB, and technical software/automation systems. Academic depth spans quantum information and computing, quantum communication and photonics, semiconductor/device technologies, and information/coding theory; project experience also includes embedded systems and industry-university engineering.

EDUCATION

M.Sc. Quantum Technologies in Electrical and Computer Engineering

Technische Universität Braunschweig, Germany | 66/120 ECTS completed

Academic focus: Quantum Information & Computing • Quantum Communication & Photonics • Semiconductor & Quantum Devices • Information & Coding Theory

Selected academic work: QKD satellite communication • Nonlinear photonics analytical coursework • Coding-theory computational work • Semiconductor crystal defects

Oct 2024 - Present

2 yrs 1 mo

Next: Internship & Thesis

B.Sc. Computer Engineering (AI specialization)

Azad University of Mashhad, Iran

Academic focus: Software, Programming & Algorithms • AI, Machine Learning & Computer Vision • Embedded Systems & Hardware • Databases & Information Systems

Sep 2019 - Feb 2024

4 yrs 6 mos

EXPERIENCE

Iran Khodro (IKCO) — Iran's largest automaker

Pouyesh Skill Development Program (Co-op)

Apr 2023 - Sep 2023

6 mos

Top 1

- **Program & selection:** Joined a university-industry initiative built around practical industry exposure and talent development; ranked 1st among 50 students from Azad University and Ferdowsi University of Mashhad after multiple competitive stages.
- **Applied computer-vision project:** Developed a MATLAB day/night vehicle-detection system for road-intersection video, addressing the challenge of substantially different daytime and nighttime illumination within one application.
- **Outcome:** Received an invitation to join Iran Khodro's R&D team following the project and chose to continue the academic path.

Ferdowsi University of Mashhad, Iran

Project Contributor - Anodyne (University-Based Startup)

Mar 2023 - Aug 2023

6 mos

- **Project coordination & external engagement:** Supported presentations, stakeholder communication, and outreach for a dental imaging initiative.
- **Industry-event delivery:** Assisted with on-site project presentation, attendee coordination, promotional activities, and professional networking during dental and medical industry events.

Azad University of Mashhad, Iran

Teaching Assistant - Theory of Formal Languages and Automata

Sep 2023 - Jan 2024

5 mos

- **Teaching & assessment:** Led weekly exercise/problem-solving sessions for approximately 20 students and evaluated assignments and exams under course supervision.

Volunteer IT Support

Oct 2022 - Aug 2023

11 mos

- **On-site technical support:** Continued voluntarily after the internship, progressing from central IT support to troubleshooting and repair across university departments.

IT Support Intern

Jul 2022 - Sep 2022

3 mos

- **Hardware & network support:** Diagnosed/repaired desktops, replaced or upgraded components, configured Windows, and supported racks/cabling, LAN, and Wi-Fi.

TECHNICAL SKILLS

Programming

Python; C; C++; MATLAB

Machine Learning & Computer Vision

PyTorch; scikit-learn

Scientific Computing & Signal Processing

NumPy; SciPy; pandas; Matplotlib

Embedded Systems

ATmega32

Development Tools & Databases

Git; GitHub; GitLab; SQL

SELECTED TECHNICAL PROJECTS

AI SYSTEMS & WORKFLOW AUTOMATION

2026

Career OS - Privacy-First AI Context & Knowledge System ([Link](#))

Google Apps Script; Google Drive API; Gemini; Groq; GitHub

- Designed and co-developed a privacy-first system for shared AI context, evidence tracking, source ingestion, provenance-aware synthesis, queue/retry processing, and CI privacy checks.

MACHINE LEARNING & COMPUTER VISION

2025

MRI Segmentation Inference Dashboard ([Link](#))

PyTorch; Streamlit

- Integrated pretrained segmentation inference with preprocessing, weight loading, masks, overlays, and area visualization.

Gas Classification Pipeline and Simulation Dashboard ([Link](#))

2025

Python; scikit-learn; RandomForest; Streamlit

- Built statistical sensor-window features and a six-gas classification dashboard with class-probability output.

Activity Recognition ML Comparison ([Link](#))

2025

Python; PyTorch; scikit-learn; Poetry

- Compared RandomForest and 1D-CNN approaches for sensor activity recognition and configured lint/test quality checks.

Day/Night Vehicle Detection ([Link](#))

2023

MATLAB; Computer Vision; Image Processing

- Developed a road-intersection pipeline with day/night routing, foreground detection, morphology, and region-based vehicle filtering.

SIGNAL & IMAGE PROCESSING

2023

Hybrid Image Generation ([Link](#)) | Stereo Sound Direction ([Link](#))

MATLAB; Image and Signal Processing

- Combined low/high spatial-frequency image content; separately analyzed stereo-audio window energy for dominant left/right direction.

EMBEDDED SYSTEMS

2021 - 2022

ATmega32 Digital Clock ([Link](#)) | ATmega32 Digital Safe ([Link](#))

Embedded C; ATmega32; 7-segment display; matrix keypad; LCD

- Implemented coursework projects for a multiplexed digital clock and a keypad/password digital safe.

SELECTED MASTER'S TECHNICAL WORK

Coding Theory Computational Lab

2026

MATLAB-based group work on cyclic-code structure, Hamming distance/correction radius, generator-matrix calculations, and recorded bit-error behavior versus SNR.

Information & Coding Theory

2025 - 2026

Graded analytical work on block/prefix coding, Kraft sums, relative entropy, Markov processes, typical sets, and state-dependent channel concepts.

Nonlinear Photonics

2025 - 2026

Analytical work on dispersion, phase matching, harmonic generation, nonlinear optical effects, fiber nonlinearities, and gain/threshold calculations.

Technical Presentations & Journal Club

2025 - 2026

Prepared work on state-dependent channel coding, quantum communication networks, semiconductor crystal defects, and a plasmonic-metasurface literature review.

HONORS

B.Sc. Computer Engineering: Ranked 5th of 131 students

2024 · Top 4%

Iran National University Entrance Exam (Konkour): ~10,000 of ~140,000

2018 · Top 7%

Khorasan Razavi selective-school Entrance Exam: 12 of ~27,000

2014 · Top 1%

LANGUAGES & CERTIFICATES

Languages

German - A2 course completed; exam result pending; next: B1.1

English - TOEFL iBT 95/120

Persian - Native

Training / Certificates

Sharif University of Technology - Quantum Programming Language (in progress)

2026

Ariaqanta Institute - Quantum Machine Learning

2025

Kaggle - Machine Learning; Introduction to Programming Using Python

2025

Azad University, Mashhad - Data & ML Bootcamp - 96 hours

2023

Jahad Daneshgahi, Khorasan Razavi - Python Programming - 40 hours

2021